

L9a: Introduction to Derivatives and European Option Pricing

CHEME 5660 Cornell University

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Objectives for Today

We introduce derivatives and begin our discussion of derivatives pricing. What is a payment that may never arrive worth today?

Today's concepts

- Define and classify derivatives: futures, forwards, swaps, and options, with their underlying assets and market structures.
- Analyze option payoffs and profits: payoff and profit for long and short calls and puts, and how the strike and premium affect profitability at expiration.
- Price European options contracts: the premium makes the expected net present value zero under the risk-neutral pricing rule. We compute it by Monte Carlo simulation and with the Black–Scholes–Merton formulas, and relate calls and puts with put–call parity.

Lectures and Examples

Lecture:
CHEME-5660-L9a-Lecture-IntroductionToDerivatives-BSM-Fall-2026.ipynb

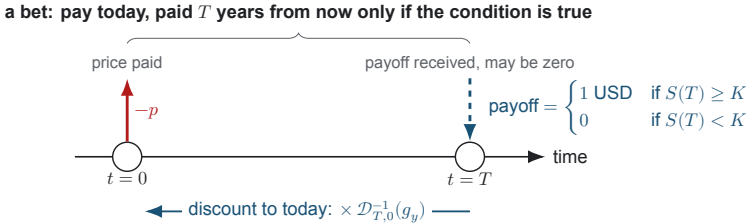
Example 1:
CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Example 2:
CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

The first example computes payoff and profit for single long and short call and put contracts. The second computes European call and put premiums by Monte Carlo simulation and with the Black–Scholes–Merton formulas, then looks at how the premium changes with strike price, in the model and in real S&P 500 index option quotes.

Concept Review: A Bet on the Future

Suppose we pay p today for a contract that pays 1 USD in T years if $S(T) \geq K$, and nothing otherwise.



Discounting the payoff at the benchmark growth rate g_y gives a random NPV:

$$\text{NPV}_0 = -p + \mathcal{D}_{T,0}^{-1}(g_y)\, V(S(T)) = \begin{cases} -p + e^{-g_y T} & \text{if } S(T) \geq K \text{ (true)} \\ -p & \text{if } S(T) < K \text{ (false)} \end{cases}$$

Expected NPV of the Bet

Under our beliefs about $S(T)$, the expected NPV is:

$$\mathbb{E}[\text{NPV}_0] = (-p + e^{-g_y T}) \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}} + (-p) \underbrace{\mathbb{P}(S(T) < K)}_{\text{false}} \qquad \text{weight by probability}$$

$$= -p \underbrace{\left[\mathbb{P}(S(T) \geq K) + \mathbb{P}(S(T) < K)\right]}_{=1} + e^{-g_y T}\, \mathbb{P}(S(T) \geq K) \qquad \text{collect the } p \text{ terms, probabilities sum to one}$$

$$= -p + e^{-g_y T} \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}}$$

The bet has a non-negative expected NPV when $p \leq e^{-g_y T}\, \mathbb{P}(S(T) \geq K)$.

The bet is a derivative contract: its payoff depends on the future price of another asset (or event). Every buyer pays one market price, yet buyers with different beliefs value the same contract differently.

How is the market price set?

Company Profile: Cboe Global Markets

Cboe operates markets for options, equities, futures, and foreign exchange. It opened on April 26, 1973, as the Chicago Board Options Exchange, the world's first exchange for listed options.

- Listed options: standard strikes, expirations, and contract sizes, so the same contract can trade many times a day. The Options Clearing Corporation (OCC) stands between the buyer and the seller.
- Index options and volatility: SPX options are European-style and cash-settled. The VIX reads the market's expected 30-day volatility of the S&P 500 from their premiums.
- Event contracts: since June 2026, binary options on the Mini-S&P 500 that pay 100 USD or nothing, cleared by OCC.

Today's question: where does a listed premium come from?

What Is a Derivative?

A derivative is a contract between a buyer and a seller to transact an asset at a set price on a future date. It is an abstract asset whose cash flows depend on an underlying asset or event.

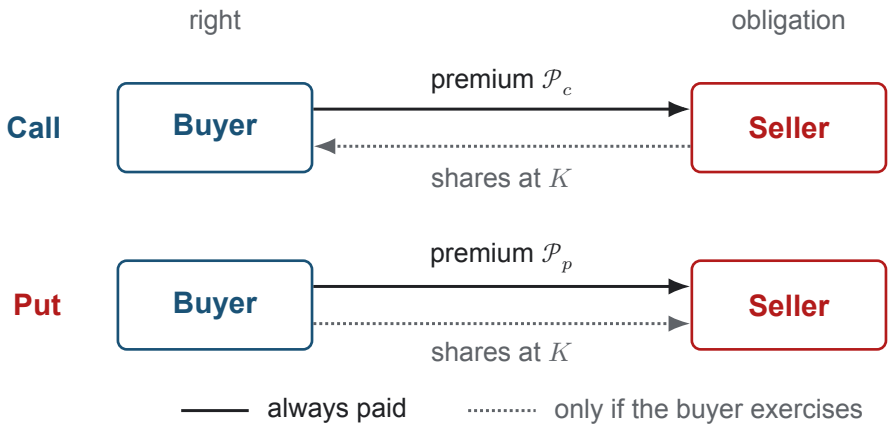
- Futures, forwards, and swaps: obligate both parties to transact. Futures trade on exchanges with standard terms. Forwards trade over the counter, so if one party cannot pay, the other takes the loss. Swaps exchange one asset's cash flows for another's, also over the counter.
- Options: the buyer holds the right, but not the obligation, to buy (call) or sell (put) at a set price. Stock, index, and ETF options are exchange-traded.
- Event contracts: pay a fixed amount, usually one dollar, if an event occurs and nothing otherwise. Kalshi lists them.

How Big Is the US Options Market?

- 2025: a record of about 15.2 billion contracts, or about 60.6 million per day (OCC).
- 2026: through September, about 13.2 billion contracts, or about 70.6 million per day (OCC).
- Notional: about 4 trillion USD per day in 2025, more than three times the 1.1 trillion USD per day traded in U.S. equities (Traders Magazine, Cboe).

Sources: the lecture notebook's market table, with dates and links.

Call and Put Options Contracts



The Business Case for Long and Short Calls

A call gives the owner the right, but not the obligation, to buy 100 shares at the strike price K by the expiration date.

- Buyer (long): gains from a rising share price without buying the shares. The most the buyer can lose is the premium.
- Seller (short): receives the premium and must sell shares at K if the call is exercised. The possible loss has no limit, so calls are often sold by investors who own the shares (a covered call).

The Business Case for Long and Short Puts

A put gives the owner the right, but not the obligation, to sell 100 shares at the strike price K by the expiration date.

- Buyer (long): gains from a falling share price without selling the shares. The most the buyer can lose is the premium.
- Seller (short): receives the premium and must buy shares at K if the put is exercised. At a share price of zero, the seller loses K minus the premium's future value per share, so puts are often sold by investors with the cash to buy the shares (a cash-secured put).

Call and Put Payoffs

The payoff per share of a single call or put contract at expiration (or when the contract is exercised) is:

$$\boxed{\begin{aligned} V_c(K, S(T)) &= \max(S(T) - K,\ 0) = (S(T) - K)^+ \\ V_p(K, S(T)) &= \max(K - S(T),\ 0) = (K - S(T))^+ \end{aligned}}$$

where K is the strike price and $S(T)$ is the share price at expiration (time $t = T$). The seller charges the buyer a premium per share, $\mathcal{P}_c$ for a call and $\mathcal{P}_p$ for a put.

Profit at Expiration

The premium is paid at $t = 0$ and the payoff arrives at $t = T$, so we compare them at expiration with the premium's future value $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$, where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$. The buyer's profit per share is:

$$P_c^{\text{buyer}}(K, S(T)) = V_c(K, S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$$
$$P_p^{\text{buyer}}(K, S(T)) = V_p(K, S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$$

With orientation $\theta = +1$ (long) or -1 (short), a position's profit is $\theta \cdot P^{\text{buyer}}$, so $P^{\text{buyer}} + P^{\text{seller}} = 0$ at every share price.

Options are a zero-sum game: one party's gain is the other's loss.

Breakeven

Setting the buyer's profit to zero gives the breakeven share price $S^{\ddagger}$ at expiration:

$$S^{\ddagger} = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c \qquad \text{call}$$

$$S^{\ddagger} = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p \qquad \text{put}$$

Above the call breakeven, and below the put breakeven, the buyer profits and the seller loses. Listed contracts often expire within weeks, so $\mathcal{D}_{T,0}(g_y)$ is close to one (about 1.003 for 30 days at $g_y = 4\%$).

Example 1: CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Pricing Put and Call Options Contracts

An option's price depends on the underlying asset's value, strike price, time to expiration, and exercise style.

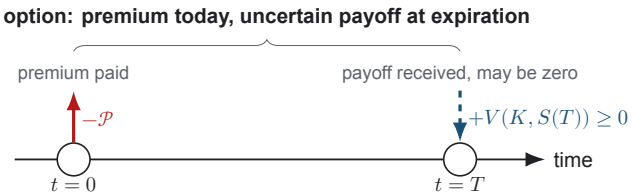
- European-style: exercise only at expiration.
- American-style: exercise any time through expiration.
- Bermudan-style: less common, exercise on specified dates.

The American option's added flexibility makes it worth at least as much as an otherwise identical European option.

Options on single stocks and on exchange-traded funds such as SPY are American-style. The European-style contracts we can trade are cash-settled index options: SPX (SPXW for weekly expirations), XSP, NDX, and RUT.

Let's start with European-style options and compute their net present value.

Options as Abstract Assets



The buyer pays the premium $\mathcal{P}$ today and receives the payoff $V(K, S(T)) \geq 0$ at expiration, so the net present value at $t = 0$ is:

$$\text{NPV}_0 = -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot V(K, S(T))$$

Factoring out $\mathcal{D}_{T,0}^{-1}(g_y)$ gives $\text{NPV}_0 = \mathcal{D}_{T,0}^{-1}(g_y) \cdot P^{\text{buyer}}$: the NPV is the buyer's profit at expiration discounted to today, so it is zero at the breakeven share price.

The European Premium

How is the market price set? A call pays $S(T) - K$ when $S(T) > K$ and nothing otherwise. Following the same steps as for the bet:

$$\begin{aligned}
\mathbb{E}[\mathrm{NPV}_0] &= \left(-\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\right)\mathbb{P}(S(T) > K) + (-\mathcal{P})\,\mathbb{P}(S(T) \le K) && \text{weighted sum} \\
&= -\mathcal{P}\ \underbrace{\left[\mathbb{P}(S(T) > K) + \mathbb{P}(S(T) \le K)\right]}_{=1} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\,\mathbb{P}(S(T) > K) && \text{collect } \mathcal{P} \text{ terms, sum to one} \\
&= -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{P}(S(T) > K)\,\bar{V}
\end{aligned}$$

where $\bar{V}$ is the average payoff when $S(T) > K$. The expected NPV is non-negative when $\mathcal{P} \le \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{P}(S(T) > K)\,\bar{V}$.

Whose Probabilities?

$\mathbb{P}$ holds our own beliefs about $S(T)$, so this is the most we would pay. The market prices every option with one shared risk-neutral pricing rule $\mathbb{Q}$, under which GBM has drift g_y. Repeating the calculation under $\mathbb{Q}$:

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}[\mathrm{NPV}_0] &= \left(-\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\right)\mathbb{Q}(S(T) > K) + (-\mathcal{P})\,\mathbb{Q}(S(T) \le K) \\
&= -\mathcal{P}\ \underbrace{\left[\mathbb{Q}(S(T) > K) + \mathbb{Q}(S(T) \le K)\right]}_{=1} \\
&\quad + \mathcal{D}_{T,0}^{-1}(g_y)\ \underbrace{\left[\bar{V}\,\mathbb{Q}(S(T) > K) + 0 \cdot \mathbb{Q}(S(T) \le K)\right]}_{=\mathbb{E}_{\mathbb{Q}}[V_c(K,S(T))]} \\
&= -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{E}_{\mathbb{Q}}\left[V_c(K, S(T))\right]
\end{aligned}$$

where $\bar{V}$ is now the average payoff when $S(T) > K$ under $\mathbb{Q}$.

The Market Premium

The second bracket is the call's expected payoff under $\mathbb{Q}$. The seller's NPV is the negative of the buyer's, so under one shared $\mathbb{Q}$, the only premium at which neither side expects to lose sets $\mathbb{E}_{\mathbb{Q}}[\mathrm{NPV}_0] = 0$:

$$\mathcal{P}_c = \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{E}_{\mathbb{Q}}[V_c(K, S(T))]$$

A put gives the same result with V_p in place of V_c. For either contract, the premium is the discounted expected payoff under $\mathbb{Q}$:

$$\boxed{\mathcal{P}^{\mathrm{EU}} = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V(K, S(T))]}$$

Event Contracts

The Concept Review bet is an event contract on $S(T) \geq K$, which pays 1 USD if the event occurs and nothing otherwise. Its market price is:

$$p = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[\mathbb{1}\{\text{event}\}] = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{Q}(\text{event})$$

The pricing probability $\mathbb{Q}(\text{event})$ can differ from the real-world probability $\mathbb{P}(\text{event})$. A general Kalshi event, such as an election, is not tied to a share price, so choosing its pricing probability requires additional assumptions.

A Monte Carlo Premium

Simulation estimates the market premium directly. The payoff depends only on $S(T)$, so we draw it from the GBM solution with drift g_y.

Initialize: given $S(0)$, K, the payoff $V(K, S)$, T (years), σ, g_y, and M paths.

1. For each path $i = 1, 2, \ldots, M$ **do**:
   a. Draw a standard normal random number Z_i.
   b. $S_i(T) \leftarrow S(0) \cdot \exp\left[(g_y - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_i\right]$.
   c. $V_i \leftarrow V(K, S_i(T))$.
2. Discount the average payoff: $\mathcal{P} \leftarrow \mathcal{D}_{T,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} V_i$.

Optional: an alternative way to think about the premium, in the contingent claims notebook.

The Black–Scholes–Merton Equation

Merton and Scholes received the 1997 Nobel Prize in Economic Sciences for this model. Fischer Black had passed away before the award.

Assume the price of a non-dividend-paying share follows GBM with constant σ and a constant risk-free rate g_y. The premium $\mathcal{P}(S,t)$, with $\mathcal{P}(S,T) = V(K,S)$, satisfies:

$$\boxed{\frac{\partial \mathcal{P}}{\partial t} + g_y S \frac{\partial \mathcal{P}}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \mathcal{P}}{\partial S^2} = g_y \mathcal{P}}$$

Only g_y and σ appear, not the real-world growth rate of the share price. Solving backward from the payoff gives $\mathcal{P}^{\text{EU}}$, the premium Monte Carlo estimates.

Derivation: the contingent claims notebook.

Black–Scholes–Merton Premiums

Instead of simulating, we can compute this premium with the closed-form European call and put formulas:

$$\boxed{\begin{aligned}\mathcal{P}_c &= N(d_+)\cdot S(0) - N(d_-)\cdot K\cdot \mathcal{D}_{T,0}^{-1}(g_y)\\ \mathcal{P}_p &= N(-d_-)\cdot K\cdot \mathcal{D}_{T,0}^{-1}(g_y) - N(-d_+)\cdot S(0)\end{aligned}}$$

where N is the standard normal CDF, T is in years, and:

$$d_+ = \frac{1}{\sigma\sqrt{T}}\left[\ln\frac{S(0)}{K} + \left(g_y + \frac{\sigma^2}{2}\right)T\right], \qquad d_- = d_+ - \sigma\sqrt{T}$$

Historical or Implied Volatility?

- Historical volatility: $\sigma_{\text{ann}} = \sqrt{\Delta t} \cdot \sigma_g$, where σ_g is the standard deviation of past growth rates and Δt is the time between observations. It looks backward.
- Implied volatility (IV): the σ "implied" by an option's market price through a model such as BSM, the market's forecast of share price moves. It is often at or above the historical volatility.
- How can we use it? IV is a one-standard-deviation move in the share price over one year. Over T years, the move scales with $\sqrt{T}$.

A Back-of-the-Envelope Range from IV

Setting $Z = \pm 1$ in the GBM solution gives a quick range for the share price:

$$
\begin{aligned}
S(T) &= S(0)\exp\left[\mu_g T + \sigma\sqrt{T}\,Z\right] && \text{solution of geometric Brownian motion} \\
S_{\pm} &= S(0)\exp\left[\mu_g T \pm \mathrm{IV}\sqrt{T}\right] && \text{set } \sigma = \mathrm{IV} \text{ and } Z = \pm 1 \\
&\approx S(0)\exp\left[\pm \mathrm{IV}\sqrt{T}\right] && \mu_g T \text{ is small next to } \mathrm{IV}\sqrt{T} \text{ for short } T \\
&\approx S(0)\left(1 \pm \mathrm{IV}\sqrt{T}\right) && e^x \approx 1 + x \text{ for small } x
\end{aligned}
$$

where μ_g is the mean growth rate and Z is standard normal.

Example: $S(0) = 100$ USD/share, $\mathrm{IV} = 40\%$, and $T = 30/365$ years give $\mathrm{IV}\sqrt{T} = 0.115$, a range of about 88.5 to 111.5 USD/share. Under the model, the share price ends inside it about 68% of the time.

Which σ?

If we knew σ, the formula would return the market premium. In practice, we run it backward. Like a bond's yield to maturity, the implied volatility $\hat{\sigma}$ is the σ at which the BSM premium matches $\mathcal{P}^{\text{market}}$.

Initialize: given $\mathcal{P}^{\text{market}}$, $S(0)$, K, T (years), g_y, a bracket $\sigma_{\text{lo}} < \sigma_{\text{hi}}$ with $\mathcal{P}(\sigma_{\text{lo}}) \le \mathcal{P}^{\text{market}} \le \mathcal{P}(\sigma_{\text{hi}})$, a tolerance $\epsilon > 0$, `maxiter`, and $i \leftarrow 1$.

While $\sigma_{\text{hi}} - \sigma_{\text{lo}} > \epsilon$ and $i \le$ `maxiter` **do**:

1. $\sigma_{\text{mid}} \leftarrow (\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$ and compute its BSM premium $\mathcal{P}(\sigma_{\text{mid}})$.
2. If $\mathcal{P}(\sigma_{\text{mid}}) < \mathcal{P}^{\text{market}}$, **then** $\sigma_{\text{lo}} \leftarrow \sigma_{\text{mid}}$. Otherwise, $\sigma_{\text{hi}} \leftarrow \sigma_{\text{mid}}$.
3. $i \leftarrow i + 1$.

The implied volatility is the midpoint of the final bracket.

Implied Volatility in Practice

- One answer: the BSM premium rises with σ, so exactly one σ matches the market premium, and each step halves the bracket.
- Why the formula: the Monte Carlo premium changes with the random draws and costs M paths at every step.
- One per contract: each strike and expiration has its own implied volatility, although BSM assumes one constant σ. A surface fitted to today's quotes predicts it for unquoted contracts or future dates (Sun et al., 2026).

Optional: the SPXW volatility skew notebook applies this bisection to a chain of S&P 500 index options.

Put–Call Parity for European Contracts

A European call and put with the same strike and expiration have payoffs that differ by $S(T) - K$. Subtracting the premiums:

$$
\begin{aligned}
\mathcal{P}_c - \mathcal{P}_p &= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_c(K, S(T))] && \text{premium formula} \\
&\quad - \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_p(K, S(T))] \\
&= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_c(K, S(T)) - V_p(K, S(T))] && \text{expectation is linear} \\
&= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[S(T) - K] && \text{payoff identity} \\
&= \mathcal{D}_{T,0}^{-1}(g_y) \cdot (S(0)\, e^{g_y T} - K) && \text{expected share price grows at } g_y \\
&= S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y) && \mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}
\end{aligned}
$$

This is put–call parity: $\boxed{\mathcal{P}_c - \mathcal{P}_p = S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y)}$

Parity holds only for European contracts. An American holder can exercise early, so the call–put difference is bounded, not fixed.

Example 2: CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

What About American-Style Options?

The holder can exercise at any time $\tau \leq T$, chosen using only the information available at that time. The premium is the largest expected payoff, discounted from τ back to today, over all such exercise times:

$$\mathcal{P}^{\mathsf{AM}} = \max_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\left[\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V(K, S(\tau))\right] \geq \mathcal{P}^{\mathsf{EU}}$$

The American put has no closed-form formula like BSM, so its premium is usually computed on a binomial lattice.

Optional Advanced Material

Contingent claims: advanced/contingent_claims/
CHEME-5660-L9a-Advanced-ContingentClaims-BSM-Fall-2026.ipynb

We derive where the BSM premiums come from: the risk-neutral probability on a one-step lattice, state prices for event contracts, the BSM equation and its solution, and the call built from a digital claim and an asset-or-nothing claim. We also compare a client's certainty equivalent for the bet with its market price.

Volatility skew: advanced/spxw_volatility_skew/
CHEME-5660-L9a-Advanced-SPXW-Volatility-Skew-Fall-2026.ipynb

We apply the implied-volatility bisection to an end-of-day chain of SPXW options and compare the skew with the realized volatility of SPY, which tracks the index.

Notebooks: lectures/week-9/L9a/advanced, also linked from the lecture notebook.

Summary

In this lecture, we explored derivatives, a fundamental asset class used for risk management, hedging, and speculation.

Key takeaways

- Derivatives derive their value from underlying assets: common types are futures, forwards, swaps, and options, each with different characteristics, maturities, and uses.
- Option payoffs are determined by strike prices and contract orientation: payoff diagrams at expiration show profitability and risk exposure.
- The premium makes the expected NPV zero under the risk-neutral pricing rule: Monte Carlo simulation or the BSM formulas compute it for European contracts, and put–call parity links calls and puts.